**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 201 Major Exam II Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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**1.** An equation of the level curve of the function $f(x,y) = \dfrac{x}{x^2+y^2}$ that passes through the point $(1,1)$ is

(a) $2x^2 - x + 2y^2 = 0$

(b) $x^2 - 2x + y^2 = 0$

(c) $x^2 - 3x + y^2 = -1$

(d) $2x^2 - 3x + 2y^2 = 1$

(e) $x^2 + x + y^2 = 3$

**2.** The slope of the surface $f(x,y) = xe^{x^2y}$ at the point $(1,2,e^2)$ in the $x$-direction is equal to

(a) $e^2$

(b) $2e^2$

(c) $5e^2$

(d) $4e^2$

(e) $3e^2$

**3.** The domain of the function $f(x,y) = \ln(y - x + 2)$ is

(a) $\{(x,y) : y \geq x - 2\}$

(b) $\{(x,y) : y > x - 2\}$

(c) $\{(x,y) : y = x - 2\}$

(d) $\{(x,y) : y < x - 2\}$

(e) $\{(x,y) : y > x - 1\}$

**4.** If $x+by+cz = d$ is an equation for the plane that passes through the points $P(0,0,0)$, $Q(1,0,1)$, and $R(-1,1,0)$, then $b + c + d =$

(a) 1

(b) $-1$

(c) 2

(d) $-2$

(e) 0

**5.** The graph of the equation $4x^2 - 3y^2 - 12z^2 = 0$ is

(a) an elliptic paraboloid

(b) a hyperboloid of two sheets

(c) an elliptic cone

(d) a hyperboloid of one sheet

(e) an ellipsoid

**6.** $\lim_{(x,y)\to(0,0)} \frac{\ln(2x^2+2y^2+1)}{x^2+y^2} =$

(a) 2

(b) 0

(c) $\frac{1}{2}$

(d) 1

(e) $\frac{1}{4}$

**7.** Let $z = \sqrt{29 - x^2 - y^2}$. If $(x, y)$ moves from the point $(3, 4)$ to the point $(3.01, 3.97)$, then the change in $z$ is approximately equal to

(a) $\frac{1}{25}$

(b) $\frac{1}{20}$

(c) $\frac{3}{25}$

(d) $\frac{7}{200}$

(e) $\frac{9}{200}$

**8.** A set of parametric equations for the line passing through the point $(2, 3, 4)$ and is perpendicular to the plane given by $3x + 2y - z = 6$ is given by

(a) $x = 2 + 3t,\ y = 3 + 2t,\ z = 4 - t$

(b) $x = 2 + 2t,\ y = 3 + 3t,\ z = 4 + 4t$

(c) $x = 1 + t,\ y = -1 + t,\ z = 2 + t$

(d) $x = 3 + 2t,\ y = 2 + 3t,\ z = -1 + 4t$

(e) $x = 2 - t,\ y = 3 + 2t,\ z = 4 + 3t$

**9.** Let $w = xy^2 + yz^2 + zx^2$, $x = s + t^2$, $y = \dfrac{s}{t}$, $z = s^2$. The value of $\dfrac{\partial w}{\partial t}$ when $s = 1,\ t = 1$ is equal to

(a) 4

(b) 0

(c) 8

(d) 5

(e) 12

**10.** The distance between the point $Q(1, 1, 1)$ and the line $L:\ x = t,\ y = -t,\ z = 2t$ is equal to

(a) $\dfrac{\sqrt{21}}{2}$

(b) $\dfrac{\sqrt{7}}{4}$

(c) $\dfrac{\sqrt{21}}{4}$

(d) $\dfrac{\sqrt{21}}{3}$

(e) $\dfrac{\sqrt{7}}{2}$

**11.** The set of all points equidistant from the point $(0, 0, 3)$ and the plane $z = -1$ forms

(a) a hyperboloid of one sheet

(b) an ellipsoid

(c) a hyperbolic paraboloid

(d) an elliptic paraboloid

(e) an elliptic cone

**12.** If $(a, b, c)$ is the point of intersection between the plane $x + 3y + z = 6$ and the line $x = t,\ y = 1 - t,\ z = 2 + 3t$, then $a + b + c =$

(a) 8

(b) $-7$

(c) 5

(d) $-4$

(e) 6

**13.** Which one of the following statements is TRUE:

$$f(x,y) = \frac{x^2 y}{x^4 + y^2}$$

(a) $\lim_{(x,y)\to(0,0)} f(x,y) = \frac{1}{2}$

(b) along the parabola $y = x^2$, $\lim_{(x,y)\to(0,0)} f(x,y) = 2$

(c) $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

(d) along the line $y = 2x$, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

(e) along the line $y = -x$, $\lim_{(x,y)\to(0,0)} f(x,y) = -1$

**14.** If $z = e^{-t}\cos\left(\frac{x}{2}\right)$, then $\frac{\partial z}{\partial t} - 4\frac{\partial^2 z}{\partial x^2} =$

(a) $2e^{-t}\sin\left(\frac{x}{2}\right)$

(b) $2e^{-t}\cos\left(\frac{x}{2}\right)$

(c) $0$

(d) $-2e^{-t}\cos\left(\frac{x}{2}\right)$

(e) $-2e^{-t}\sin\left(\frac{x}{2}\right)$

**15.** If $z$ is defined implicitly as a differentiable function of $x$ and $y$ by the equation

$$xyz = \sin(3x - 2y - z),$$

then $\frac{\partial z}{\partial x} =$

(a) $\dfrac{\cos(3x-2y-z) + yz}{3\cos(3x-2y-z) - xy}$

(b) $\dfrac{3\cos(3x-2y-z) + xy}{\cos(3x-2y-z) - yz}$

(c) $\dfrac{3yz - \cos(3x-2y-z)}{xy + \cos(3x-2y-z)}$

(d) $\dfrac{3\cos(3x-2y-z) - yz}{\cos(3x-2y-z) + xy}$

(e) $\dfrac{3\sin(3x-2y-z) - yz}{\sin(3x-2y-z) + xy}$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(b)** | $f(1,1)=\frac{1}{2}$, so $2x=x^2+y^2$, i.e. $x^2-2x+y^2=0$. |
| 2 | **(c)** | $f_x=2x^2ye^{x^2y}+e^{x^2y}$, so $f_x(1,2)=4e^2+e^2=5e^2$. |
| 3 | **(b)** | Need $y-x+2>0$, i.e. $y>x-2$. |
| 4 | **(e)** | $\mathbf{n}=\overrightarrow{PQ}\times\overrightarrow{PR}=\langle -1,-1,1\rangle$, plane $x+y-z=0$: $b+c+d=0$. |
| 5 | **(c)** | $4x^2=3y^2+12z^2$ has the form of an elliptic cone (axis the $x$-axis). |
| 6 | **(a)** | With $u=x^2+y^2$: $\lim\frac{\ln(2u+1)}{u}=\lim\frac{2}{2u+1}=2$. |
| 7 | **(e)** | $dz=-\frac{3}{2}(0.01)-2(-0.03)=0.045=\frac{9}{200}$. |
| 8 | **(a)** | Direction = normal $\langle 3,2,-1\rangle$: $x=2+3t,\ y=3+2t,\ z=4-t$. |
| 9 | **(d)** | $w_xx_t+w_yy_t=5\cdot 2+5\cdot(-1)=5$ at $s=t=1$. |
| 10 | **(d)** | $D=\frac{\|\langle 3,-1,-2\rangle\|}{\|\langle 1,-1,2\rangle\|}=\frac{\sqrt{14}}{\sqrt{6}}=\frac{\sqrt{21}}{3}$. |
| 11 | **(d)** | $\sqrt{x^2+y^2+(z-3)^2}=\|z+1\|$ gives $z=\frac{x^2+y^2}{8}+1$: elliptic paraboloid. |
| 12 | **(e)** | $t+3(1-t)+2+3t=6\Rightarrow t=1$, point $(1,0,5)$, sum 6. |
| 13 | **(d)** | Along $y=2x$: $\frac{2x}{x^2+4}\to 0$; the limit along $y=x^2$ is $\frac{1}{2}$, so no overall limit. |
| 14 | **(c)** | $z_t=-e^{-t}\cos\frac{x}{2}$, $z_{xx}=-\frac{1}{4}e^{-t}\cos\frac{x}{2}$, so the difference is 0. |
| 15 | **(d)** | $F=\sin(3x-2y-z)-xyz$, $z_x=-F_x/F_z=\frac{3\cos(\cdot)-yz}{\cos(\cdot)+xy}$. |